

Uttkarsh Thakur

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SKILLS

Languages: Java, Python, TypeScript, SQL, Bash, PHP, C++

Backend: Spring Boot, Spring Web MVC, Spring Data JPA, Hibernate, REST API Design, Bean Validation, FastAPI, Flask

Cloud & DevOps: Microsoft Azure, Cloudflare, Linux (Ubuntu), Nginx, Docker, SSH, TLS/HTTPS, Git & GitHub

Databases: PostgreSQL, MySQL, MariaDB, H2, MongoDB, Flyway Migrations, Schema Design

Frontend: React.js, TypeScript, Tailwind CSS, Vite, HTML/CSS

Testing & Tools: JUnit 5, Mockito, AssertJ, Maven, Swagger/OpenAPI, Postman, Google Analytics

AI/ML: RAG Pipelines, Semantic Search, FAISS, LangChain, Transformers, PyTorch, TensorFlow, Scikit-learn

PROJECTS

Knowledge Park Guide | Azure, Linux, Nginx, PHP, MariaDB, Cloudflare, Python, Bash

Sep 2026 – Present

- Built and now operate knowledgeparkguide.in, a **live production website** on a **Microsoft Azure** cloud server, configuring the full **Linux, Nginx, PHP** and **MariaDB** stack by hand and tuning it to run inside 1 GB of memory at **no compute cost** by sizing it within the free tier.
- Secured the site end to end with **Cloudflare**, **HTTPS certificates** that renew themselves automatically, key-only server access, and logging that never records passwords.
- Wrote deployment and publishing tooling in **Python**, **Bash** and **PHP** that releases changes automatically, records every command run on the server as an **audit trail**, and can be re-run safely without creating duplicates.
- Engineered a custom theme measuring **99 of 100 on Google PageSpeed** at launch with a perfect **accessibility** score and zero layout shift, and diagnosed a production outage affecting shared links down to a missing database table.

BillsSync | Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker

2026

- Built a **full-stack web application** for splitting group expenses: members log what they paid, and the app works out who owes whom, backed by a **Java Spring Boot** REST API of **16 endpoints** and a **React** and **TypeScript** interface.
- Solved a rounding problem that makes shared balances drift over time, so every split adds up to the **exact total** with no lost fractions; verified automatically across thousands of amount and group-size combinations.
- Developed a **settle-up feature** that turns a tangle of debts into the fewest payments possible, reducing a 5-member group with 8 outstanding debts to just **4 transfers**.
- Designed the **PostgreSQL** database of **6 tables** with version-controlled migrations and wrote **73 automated tests** covering business rules and every API endpoint; tuned a slow screen that made **22 database calls** down to **1**.

CodeNav AI | FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython

2026

- Built an **AI-powered search tool** that lets a developer ask questions about an unfamiliar codebase in plain English and get back the relevant code, using **large language models** and **semantic search (RAG)**.
- Improved answer quality by splitting code along real function and class boundaries instead of fixed-size blocks, so results come back complete and in context.
- Cut the time to open a codebase from **over 20 seconds to under 0.1 seconds**, a **99.5% improvement**, by saving the search index to disk instead of rebuilding it each time.

EDUCATION

Bennett University

Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering | **CGPA: 8.17/10**

Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

DAV Public School

Graduated 2022

Senior Secondary Education (Class XII)

Ranchi, India

CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera

Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera

Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera

Nov. 2024

ACHIEVEMENTS

Ranked **101st nationally** at **Smart India Hackathon 2024**

CodeChef: 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)